Prac-04

You are required to create a workflow configuration file:(.GitHub/workflows/ci.yml) that perform the following tasks:

1. checks out the source code from the repo

2. Sets up the appropriate java development kit(jdk) version required for the project.

3. Install all the necessary dependencies using maven

4. Compiles the application and execute unit tests using appropriate Maven command

5. Archive the generated build artifacts(such as JAR or WAR files) for the future use.

```
C:\Users\narut>cd mavengithubaction

C:\Users\narut\mavengithubaction>
C:\Users\narut\mavengithubaction>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\mavengithubaction

09-04-2026  12:50    <DIR>          .
10-04-2026  11:56    <DIR>          ..
09-04-2026  12:51    <DIR>          mavengit
               0 File(s)              0 bytes
               3 Dir(s)   8,994,557,952 bytes free

C:\Users\narut\mavengithubaction>cd mavengit

C:\Users\narut\mavengithubaction\mavengit>cd .github\workflows
The system cannot find the path specified.

C:\Users\narut\mavengithubaction\mavengit>mkdir .github\workflows

C:\Users\narut\mavengithubaction\mavengit>cd .github\workflows

C:\Users\narut\mavengithubaction\mavengit\.github\workflows>notepad ci-cd.yml

C:\Users\narut\mavengithubaction\mavengit\.github\workflows>cd ..

C:\Users\narut\mavengithubaction\mavengit\.github>cd ..

C:\Users\narut\mavengithubaction\mavengit>git init
'git' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\narut\mavengithubaction\mavengit>
```

STEP-01: Create the folder and buils the template

STEP-02: Get inside the folder

STEP-03: Make the directory .github/workflows

STEP-04: Create the file named ci-cd.yml with the content in it.

```
MINGW64:/c/Users/narut/mavengithubaction/mavengit
User@Ashika MINGW64 ~ (master)
$ cd mavengithubaction

User@Ashika MINGW64 ~/mavengithubaction (master)
$ git branch main
fatal: not a valid object name: 'master'

User@Ashika MINGW64 ~/mavengithubaction (master)
$ cd mavengit

User@Ashika MINGW64 ~/mavengithubaction/mavengit (master)
$ git init
Initialized empty Git repository in C:/Users/narut/mavengithubaction/mavengit/.git/

User@Ashika MINGW64 ~/mavengithubaction/mavengit (master)
$ git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-com.example.AppTest.xml', LF will be replaced by CRLF the next time Git touches it

User@Ashika MINGW64 ~/mavengithubaction/mavengit (master)
$ git commit -m "Maven workflow"
[master (root-commit) 98471ca] Maven workflow
 14 files changed, 193 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/mavengit-1.0-SNAPSHOT.jar
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class

User@Ashika MINGW64 ~/mavengithubaction/mavengit (master)
$ git remote add origin https://github.com/Ashika07734/Github-workflow-maven.git

User@Ashika MINGW64 ~/mavengithubaction/mavengit (master)
$ git branch -M main

User@Ashika MINGW64 ~/mavengithubaction/mavengit (main)
$ git push origin main
Enumerating objects: 42, done.
Counting objects: 100% (42/42), done.
Delta compression using up to 12 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (42/42), 6.68 KiB | 760.00 KiB/s, done.
Total 42 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ashika07734/Github-workflow-maven.git
 * [new branch]      main -> main

User@Ashika MINGW64 ~/mavengithubaction/mavengit (main)
$
```

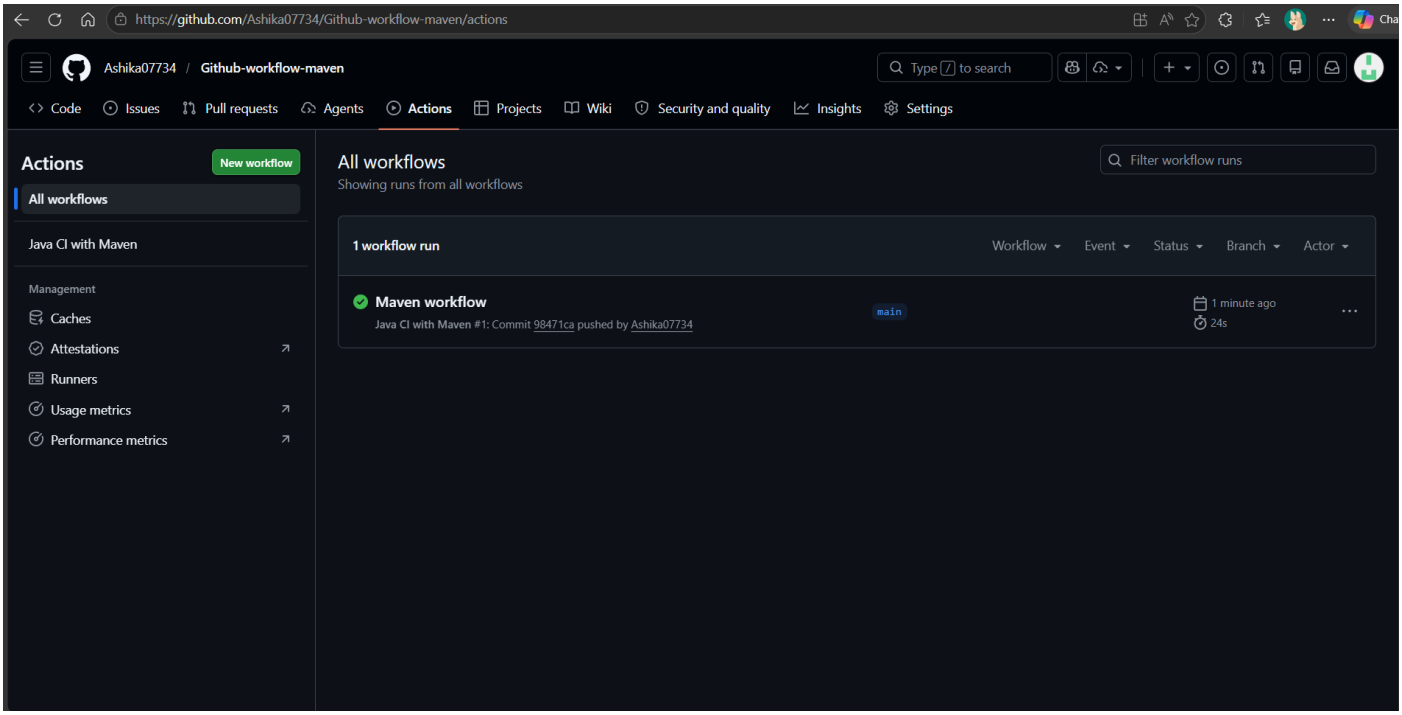
STEP-05: Git init the folder

git add .

Git commit -m “ message”

Git remote origin <url of git hub repo>

Git push origin main



STEP-06: Visit the repo go to the Actions Tab and see the workflow is running.